Elementary Analysis

Section 22: More on Metric Spaces: Connectedness

Exercises 22.1–22.14, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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22 More on metric spaces: connectedness

A separation of a set is a partition into two nonempty relatively open sets. A set is connected when it has no separation. A path is a continuous map from an interval; path-connectedness asks for a path between every two points of the set. Path-connected sets are connected, but the converse fails. In this section $C(S)$ means the *bounded* continuous real-valued functions on $S \subseteq \mathbb{R}$, and its distance is

$$d(f,g) = \|f-g\|_\infty = \sup_{x\in S} |f(x)-g(x)|.$$

Here $\|h\|_\infty$ denotes the supremum of $|h|$. When S is empty, there is just one function and its distance from itself is taken to be zero; the nontrivial arguments below concern nonempty S.

22.1 Exercise 22.1: Two impossible continuous surjections

(a) A disconnected union of intervals.

The interval $[0,1]$ is connected. By Theorem 22.2 a continuous image must be connected. But $[0,1]\cup[2,3]$ has a separation into its two nonempty pieces, obtained by intersecting with $(-\infty, 3/2)$ and $(3/2,\infty)$. Hence no continuous map from $[0,1]$ is onto that union.

(b) A rational range.

The interval $(0,1)$ is connected, whereas $\mathbb{Q}$ is not an interval: for example, it contains $1, 2$ but not $\sqrt{2}$ between them. Equivalently, $\mathbb{Q}\cap(-\infty,\sqrt{2})$ and $\mathbb{Q}\cap(\sqrt{2},\infty)$ separate $\mathbb{Q}$.

Conclusion. Neither surjection exists, because a continuous image of a connected set is connected.

22.2 Exercise 22.2: The unit circle is connected

What we must show. Show the set $x_1^2 + x_2^2 = 1$ is connected.

Give a continuous parametrization with exactly that image.

Define $\gamma : [0, 2\pi] \to \mathbb{R}^2$ by $\gamma(t) = (\cos t, \sin t)$. Both coordinates are continuous, so Proposition 21.2 proves continuity of γ. The identity $\cos^2 t + \sin^2 t = 1$ puts its image on the circle. Conversely every point of the unit circle has a polar angle $t \in [0, 2\pi]$, and hence is of this form.

Conclusion. The circle is the continuous image of the connected interval $[0, 2\pi]$, so Theorem 22.2 makes it connected.

22.3 Exercise 22.3: The closure of a connected set is connected

What we must show. Prove connectedness of $\overline{E}$ from connectedness of E.

Suppose there were a separation of the closure.

Let open ambient sets U, V give disjoint nonempty sets $\overline{E}\cap U$, $\overline{E}\cap V$ whose union is $\overline{E}$. Choose $u \in \overline{E}\cap U$. Since U is an open neighborhood of a closure point, $U \cap E \neq \varnothing$. The same argument gives $V \cap E \neq \varnothing$.

Restrict the separation.

The sets $E \cap U$ and $E \cap V$ are disjoint, nonempty and relatively open in E, and their union is E. They would separate E, contrary to its connectedness.

Conclusion. No separation of $\overline{E}$ exists. For $E = \varnothing$, the closure is empty and connected by definition.

22.4 Exercise 22.4: The closed topologist's sine curve

What we must show. Study $E = \{(x, \sin(1/x)) : 0 < x \leq 1\}$.

(a) Find all new limit points.

We claim

$$\overline{E} = E \cup (\{0\} \times [-1, 1]).$$

If $(x_n, \sin(1/x_n)) \to (a, b)$, then $a \in [0, 1]$ and $b \in [-1, 1]$. If $a > 0$, continuity of $\sin(1/x)$ at a forces $b = \sin(1/a)$, so the limit is already in E. If $a = 0$, it lies on the stated vertical segment. Conversely, given $b \in [-1, 1]$, choose an angle θ with $\sin\theta = b$. For all sufficiently large integers n, $x_n = 1/(\theta + 2\pi n)$ belongs to $(0, 1]$ and tends to zero, while $\sin(1/x_n) = b$. Thus $(0, b) \in \overline{E}$.

(b) Establish connectedness before dealing with paths.

The map $x \mapsto (x, \sin(1/x))$ is continuous on the connected interval $(0, 1]$. Its image E is connected by Theorem 22.2, and Exercise 22.3 makes its closure connected.

(c) Explain why no path can leave the vertical segment.

Suppose a path $\gamma(t) = (u(t), v(t))$, $0 \leq t \leq 1$, starts on the vertical segment and ends in E. The coordinate functions are continuous, $u(0) = 0$, and $u(1) > 0$. The zero set of u is a nonempty closed subset of $[0, 1]$, so it has a largest element $t_0 < 1$. We have $u(t_0) = 0$ and $u(t) > 0$ for $t > t_0$.

Choose a fixed $t_1 > t_0$. Because $u(t_1) > 0$, the intermediate value theorem applied to u on $[t_0, t_1]$ shows that it attains every sufficiently small positive number. In particular, it attains the numbers

$$a_n = \frac{1}{\pi/2 + 2\pi n}, \qquad b_n = \frac{1}{3\pi/2 + 2\pi n}.$$

Choose corresponding parameters $r_n, s_n \in (t_0, t_1]$. They tend to t_0: for each $\eta > 0$ with $t_0 + \eta \leq t_1$, the positive continuous u has a positive minimum on $[t_0 + \eta, t_1]$; eventually a_n, b_n are smaller than that minimum. Thus their preimages cannot lie in this latter interval.

For positive first coordinate, membership in $\overline{E}$ forces $v(t) = \sin(1/u(t))$. Hence $v(r_n) = 1$ and $v(s_n) = -1$. Both parameter sequences approach t_0, contradicting continuity of v there.

Conclusion. The closure is connected but is not path-connected. An accumulating curve can touch a segment in its closure without supplying any continuous path from that segment to the curve.

22.5 Exercise 22.5: Unions and intersections of connected sets

(a) A shared point joins two connected sets.

Suppose $E \cap F \neq \varnothing$ but $E \cup F$ has a separation A, B. Restricting the separation to E shows E lies entirely in one side, since otherwise it would be disconnected. Similarly, F lies entirely in one

side. A point of $E \cap F$ forces them to lie on the same side. The other side would then be empty, a contradiction.

(b) An intersection can be disconnected.

Let

$$E = \{(x, y) : x^2 + y^2 = 1,\ y \geq 0\}, \qquad F = \{(x, y) : x^2 + y^2 = 1,\ y \leq 0\}.$$

Each semicircle is a continuous image of a closed interval, hence connected. But $E \cap F = \{(-1, 0), (1, 0)\}$ has two isolated points and is disconnected: balls of radius $1/2$ about these points separate it.

Conclusion. A nonempty intersection guarantees a connected union; it does not guarantee a connected intersection.

22.6 Exercise 22.6: Why the supremum distance is a metric on bounded functions

(a) Verify finiteness and the metric axioms.

If $|f| \leq M$ and $|g| \leq N$ on S, then $0 \leq |f(x) - g(x)| \leq M + N$. Thus $d(f, g)$ is finite and nonnegative. It is symmetric because $|f(x) - g(x)| = |g(x) - f(x)|$. If $d(f, g) = 0$, then $|f(x) - g(x)| = 0$ at every x, so $f = g$; conversely identical functions have distance zero. Finally for every x,

$$|f(x) - h(x)| \leq |f(x) - g(x)| + |g(x) - h(x)| \leq d(f, g) + d(g, h).$$

Taking the supremum over x proves the triangle inequality.

(b) Distinguish a finite metric from a convergence definition.

Without boundedness, a pair such as $f(x) = x$, $g(x) = 0$ on $\mathbb{R}$ would have $d(f, g) = \infty$, forbidden for an ordinary metric. Uniform convergence itself only asks whether the errors $|f_n(x) - f(x)|$ are eventually small for all x simultaneously. The functions can be unbounded while their differences are uniformly small: $f_n(x) = x + 1/n$ converges uniformly to $f(x) = x$ on $\mathbb{R}$.

Conclusion. Boundedness makes the supremum distance finite for every pair in $C(S)$; it is not needed to state uniform convergence.

22.7 Exercise 22.7: A bounded sequence is a bounded continuous function on $\mathbb{N}$

What we must show. Identify the metric space B from Exercise 13.3 with $C(\mathbb{N})$.

Use discreteness to understand continuity.

Under the usual distance inherited from $\mathbb{R}$, the ball of radius $1/2$ about $n \in \mathbb{N}$ contains only n. Hence *every* function $f : \mathbb{N} \to \mathbb{R}$ is continuous: choose $\delta = 1/2$ at each point, and all tested values are identical to the center value.

Match both the elements and their distances.

The sequence $(x_1, x_2, \ldots)$ corresponds to the function $f(n) = x_n$. A sequence is bounded exactly when its function is bounded. Moreover,

$$\sup_{n \in \mathbb{N}} |x_n - y_n| = \sup_{n \in \mathbb{N}} |f(n) - g(n)|.$$

Conclusion. This correspondence is a bijection preserving distance, so B and $C(\mathbb{N})$ are the same metric space after relabeling their elements.

22.8 Exercise 22.8: Evaluation at one point is uniformly continuous

What we must show. For fixed $s_0 \in S$, study $F(f) = f(s_0)$.

Bound one error by the largest error.

For $f, g \in C(S)$,

$$|F(f) - F(g)| = |f(s_0) - g(s_0)| \leq \sup_{x \in S} |f(x) - g(x)| = d(f, g).$$

Given $\varepsilon > 0$, take $\delta = \varepsilon$. If $d(f, g) < \delta$, the displayed inequality gives $|F(f) - F(g)| < \varepsilon$.

Conclusion. Evaluation is 1-Lipschitz and therefore uniformly continuous.

22.9 Exercise 22.9: A straight line in a function space

What we must show. Prove uniform continuity of $F(t) = tf + (1 - t)g$ as a map $\mathbb{R} \to C(S)$.

First check that the values belong to the claimed space.

For fixed t, the function $tf + (1 - t)g$ is continuous and is bounded by $|t|\|f\|_\infty + |1 - t|\|g\|_\infty$. Thus $F(t) \in C(S)$.

Compute the distance between two values.

The difference is $F(t) - F(s) = (t - s)(f - g)$, so

$$d(F(t), F(s)) = |t - s| \, \|f - g\|_\infty.$$

Let $M = \|f - g\|_\infty$. If $M = 0$, F is constant. Otherwise $\delta = \varepsilon/M$ makes $|t - s| < \delta$ imply $d(F(t), F(s)) < \varepsilon$.

Conclusion. The map is Lipschitz with constant $\|f - g\|_\infty$. The estimate works on the whole real line.

22.10 Exercise 22.10: Translation is continuous in the supremum metric

What we must show. For bounded uniformly continuous $f : \mathbb{R} \to \mathbb{R}$, prove uniform continuity of $F(x) = f_x$, where $f_x(y) = f(x + y)$.

Check each translated function.

Translation preserves continuity, and $|f_x(y)| \leq \|f\|_\infty$. Thus $f_x \in C(\mathbb{R})$ for every real x.

Keep the error bound uniform in the variable being translated.

Fix $\varepsilon > 0$. Uniform continuity of f supplies $\delta > 0$ such that $|u - v| < \delta$ implies $|f(u) - f(v)| < \varepsilon/2$ for all real u, v. If $|x - z| < \delta$, then for *every* y,

$$|f_x(y) - f_z(y)| = |f(x + y) - f(z + y)| < \varepsilon/2,$$

because $|(x+y)-(z+y)| = |x-z|$. Taking a supremum yields $d(F(x), F(z)) \leq \varepsilon/2 < \varepsilon$.

Conclusion. The same δ works for all x, z, proving uniform continuity. Using $\varepsilon/2$ avoids losing the strict inequality at a supremum.

22.11 Exercise 22.11: The closed unit ball and connectedness in $C(S)$

What we must show. Let $\mathcal{E} = \{f \in C(S) : \|f\|_\infty \leq 1\}$.

(a) Its complement is open.

Suppose $f \notin \mathcal{E}$. Then some $x_0 \in S$ has $|f(x_0)| > 1$; otherwise 1 would be an upper bound for all $|f(x)|$. Put $r = (|f(x_0)| - 1)/2 > 0$. If $d(f, g) < r$, then

$$|g(x_0)| \geq |f(x_0)| - |g(x_0) - f(x_0)| > |f(x_0)| - r > 1.$$

Thus every g in this metric ball is also outside $\mathcal{E}$. The complement is open, so $\mathcal{E}$ is closed.

(b) The entire space is path-connected.

For any $f, g \in C(S)$, Exercise 22.9 gives a continuous path $t \mapsto (1-t)f + tg$, $0 \leq t \leq 1$, from f to g. Thus $C(S)$ is path-connected, and Theorem 22.5 makes it connected.

(c) The same paths stay in the unit ball.

If $f, g \in \mathcal{E}$ and $0 \leq t \leq 1$, then at every x,

$$|(1-t)f(x) + tg(x)| \leq (1-t)|f(x)| + t|g(x)| \leq 1.$$

The path therefore lies in $\mathcal{E}$.

Conclusion. $\mathcal{E}$ is closed and path-connected, hence connected; $C(S)$ is connected as well.

22.12 Exercise 22.12: A topology determined by finitely many evaluations

What we must show. Verify the topology axioms for the specified pointwise neighborhoods. Write

$$N(f; F, \varepsilon) = \{g \in C(S) : |g(x) - f(x)| < \varepsilon \text{ for all } x \in F\},$$

where $F \subseteq S$ is finite and $\varepsilon > 0$.

(a) Recall what must be proved.

Discussion 13.7 describes the characteristic properties of open sets: the empty set and the entire space are open, arbitrary unions of open sets are open, and finite intersections of open sets are open. These properties define a topology even when no metric has been specified. Here a set U is declared open if each $f \in U$ has some $N(f; F, \varepsilon) \subseteq U$.

(b) Check all three properties.

The empty set is open because it has no points for which the requirement could fail. The whole space is open since every such neighborhood lies in it (one can take $F = \varnothing$). If $f \in \bigcup_\alpha U_\alpha$, then f belongs to some open U_{α_0}, which supplies a neighborhood contained in U_{α_0} and hence in the union.

For a finite intersection $U_1 \cap \cdots \cap U_m$ and a point f in it, choose $N(f; F_j, \varepsilon_j) \subseteq U_j$ for every j. The set $F = F_1 \cup \cdots \cup F_m$ is finite, and $\varepsilon = \min_j \varepsilon_j > 0$. A function in $N(f; F, \varepsilon)$ satisfies every constraint for every U_j, so this neighborhood lies in the intersection. The empty finite intersection is the whole space, already covered.

Why these are pointwise rather than uniform neighborhoods.

Each neighborhood restricts only finitely many values of a function. A small supremum-metric ball restricts values at *every* point. For example, on $S = \mathbb{R}$ any finite set of prescribed points misses some x_0; a continuous bounded bump supported near x_0 can be zero at every prescribed point yet have supremum 2. Thus no finite-point neighborhood of zero is contained in the supremum ball of radius 1.

Conclusion. The declared sets satisfy every topology axiom. The text's additional blanket remark about nonmetrizability needs qualification: for finite S this is the usual finite-dimensional topology, and for countable $S = \{s_1, s_2, \ldots\}$ it is metrized by $\sum_{j \geq 1} 2^{-j} \min(1, |f(s_j) - g(s_j)|)$. The requested topology verification is valid for every S.

22.13 Exercise 22.13: A connected graph with an oscillatory endpoint

What we must show. Analyze the graph with the oscillating branch at positive x and the added origin.

Clarify the printed notation.

The exercise prints $f(x) = (x, \sin(1/x))$ for $x > 0$ and $f(0) = 0$. Read literally, this is a map to $\mathbb{R}^2$ with $f(0) = (0, 0)$, and its graph lies in $\mathbb{R}^3$. The frequently intended scalar formulation is $f(x) = \sin(1/x)$, $f(0) = 0$. Both give the same connectedness answer: the literal graph is homeomorphic to

$$H = \{(x, \sin(1/x)) : x > 0\} \cup \{(0, 0)\}$$

by the map $(x, y) \mapsto (x, x, y)$, whose inverse on its image is $(a, b, c) \mapsto (a, c)$. Both maps are continuous coordinate maps. Thus we prove the assertions for H.

The added origin does not disconnect the branch.

The branch $A = \{(x, \sin(1/x)) : x > 0\}$ is a continuous image of the interval $(0, \infty)$, hence connected. Also $(0, 0) \in \overline{A}$, since $(1/(n\pi), 0) \to (0, 0)$. If H had a separation, connectedness would put all of A on one side. The other side would consist of the origin and be relatively open in H, which would provide a neighborhood of the origin missing A. This contradicts its being a closure point. Hence H is connected.

It has no path from the origin to the branch.

The proof of Exercise 22.4(c) applies without change. A proposed path has a last parameter t_0 at which its first coordinate is zero. Immediately after t_0 its first coordinate is positive and tends to zero. It must attain the values a_n and b_n from that exercise at parameters tending to t_0, forcing its second coordinate to take both 1 and -1 arbitrarily close to t_0. That contradicts continuity. The restriction $x \leq 1$ was not used in that argument.

Continuity of the original function fails at zero.

Along $x_n = 1/(\pi/2 + 2\pi n)$, the sine coordinate is 1, not tending to its value 0 at zero. The sequential criterion therefore rules out continuity at zero in either formulation; continuity holds for $x > 0$.

Conclusion. The graph is connected but not path-connected, and the function is not continuous on $[0, \infty)$.

22.14 Exercise 22.14: Small successive changes force an interval of cluster values

What we must show. For bounded (s_n) with $s_{n+1} - s_n \to 0$, show its set S of subsequential limits is a connected interval.

Plan. If the sequence repeatedly visits both sides of a level c, it must cross c repeatedly. Small jumps make some crossing terms arbitrarily close to c.

Fix any level between two subsequential limits.

Take $a, b \in S$ with $a < c < b$. For each integer $j \geq 1$, choose N_j large enough that $|s_{n+1} - s_n| < 1/j$ for all $n \geq N_j$. Since a and b are subsequential limits, there are arbitrarily late terms below c and arbitrarily late terms above c. Indeed, use errors smaller than $(c-a)/2$ and $(b-c)/2$ in their defining subsequences.

Locate crossings farther and farther out.

Choose $p_j > \max(N_j, q_{j-1})$ with $s_{p_j} < c$, and then choose $q_j > p_j$ with $s_{q_j} > c$ (start with $q_0 = 0$). Let k_j be the first index after p_j for which $s_{k_j} \geq c$; it exists because q_j is one such index. Then

$$s_{k_j-1} < c \leq s_{k_j}, \qquad 0 \leq s_{k_j} - c < s_{k_j} - s_{k_j-1} < 1/j.$$

The chosen indices strictly increase, so (s_{k_j}) is a subsequence and converges to c. Thus $c \in S$.

Conclude connectedness.

The preceding argument shows S contains every number strictly between any two of its points, which is exactly the interval property. Boundedness and Bolzano–Weierstrass ensure S is nonempty; the cluster-set theorem also makes it closed and bounded. Thus it is a closed bounded interval, possibly a singleton, and intervals are connected.

Conclusion. The cluster values cannot have a gap because vanishing step sizes prevent repeated crossings from jumping across such a gap.